

32.2 The linear mixed model

The one-way random effects model of [Section 32.1](#) has one random factor and no regressors. Real studies have both: a trial has fixed treatment effects and random patient effects, a panel has fixed economic regressors and random firm effects, a field trial has fixed varieties and random blocks. The model that holds them all keeps the usual fixed part and adds a second design matrix whose coefficients are random.

32.2.1 Definition and the marginal model

Definition 32.2.1. Linear mixed model

Let $\mathbf{X}$ be a known $n \times p$ matrix and $\mathbf{Z}$ a known $n \times q$ matrix. The **linear mixed model** is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}, \quad (32.2.1)$$

where $\boldsymbol{\beta} \in \mathbb{R}^p$ is a vector of unknown constants, the **fixed effects**; $\mathbf{u}$ is a $q \times 1$ random vector, the **random effects**, with $E(\mathbf{u}) = \mathbf{0}$ and $\text{Cov}(\mathbf{u}) = \mathbf{G}$; $\boldsymbol{\varepsilon}$ is an $n \times 1$ random vector with $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \mathbf{R}$; and $\text{Cov}(\mathbf{u}, \boldsymbol{\varepsilon}) = \mathbf{0}$. The matrices $\mathbf{G}$ and $\mathbf{R}$ are nonnegative definite and are known functions $\mathbf{G}(\boldsymbol{\theta})$, $\mathbf{R}(\boldsymbol{\theta})$ of an unknown parameter $\boldsymbol{\theta}$, the **variance components**. The **normal** linear mixed model adds $\mathbf{u} \sim N_q(\mathbf{0}, \mathbf{G})$ independent of $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \mathbf{R})$.

Taking expectations in [Eq. 32.2.1](#) and using Theorem 2.2.1 gives the **marginal model**

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y}) = \mathbf{V}(\boldsymbol{\theta}) = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}, \quad (32.2.2)$$

and under normality $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \mathbf{V})$. Everything Part VII knows about the general Gauss–Markov model applies to [Eq. 32.2.2](#) at once. What distinguishes a mixed model from the general model of [Chapter 31](#) is that $\mathbf{V}$ is not an arbitrary unknown matrix but is built from a handful of parameters by a structure the experiment dictates. An $n \times n$ covariance has $n(n+1)/2$ free entries; a random intercept model has two.

Remark (Conditional and marginal). Two means live in [Eq. 32.2.1](#). Conditionally on the random effects, $E(\mathbf{Y} | \mathbf{u}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}$: this is what the subjects in *this* sample did. Marginally, $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$: this is what the population does on average. In the linear model with normal errors the two lead to the same $\boldsymbol{\beta}$, and the distinction looks pedantic. It is not pedantic at all once the link is nonlinear. For a generalized linear mixed model ([Chapter 40](#)) the marginal mean of $g^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u})$ is not $g^{-1}(\mathbf{X}\boldsymbol{\beta})$, and the subject-specific and population-averaged coefficients genuinely differ (Proposition 40.1.1).

Warning. The pair $(\mathbf{G}, \mathbf{R})$ is not identified by the marginal distribution; only $\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}$ is. In the one-way layout with $\mathbf{Z}$ the group indicators, the model with $\mathbf{G} = \sigma_a^2 \mathbf{I}_g$, $\mathbf{R} = \sigma^2 \mathbf{I}_n$ has exactly the same marginal distribution as the model with no random effects at all and $\mathbf{R} = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$, that is, with equicorrelated errors within groups. Every likelihood-based procedure in this chapter estimates $\mathbf{V}$, so it cannot tell the two apart. What the random-

effects reading adds is a *target for prediction*: the vector $\mathbf{u}$, whose realized value [Section 32.3](#) predicts. It also adds a constraint, since $\sigma_a^2 \geq 0$ forces a nonnegative intraclass correlation, whereas the equicorrelated model allows ρ down to $-1/(m-1)$.

32.2.2 Inverting the covariance matrix

Formulas in this chapter are written with $\mathbf{V}^{-1}$, which is $n \times n$. When q is much smaller than n , or when the data fall into independent clusters, $\mathbf{V}^{-1}$ never has to be formed.

Proposition 32.2.1. The inverse and determinant of $\mathbf{V}$

Let $\mathbf{R}$ and $\mathbf{G}$ be positive definite in [Definition 32.2.1](#). Then $\mathbf{V}$ is positive definite, and

$$\mathbf{V}^{-1} = \mathbf{R}^{-1} - \mathbf{R}^{-1}\mathbf{Z}(\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z})^{-1}\mathbf{Z}^\top \mathbf{R}^{-1}, \quad (32.2.3)$$

$$\det \mathbf{V} = \det \mathbf{R} \det \mathbf{G} \det(\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z}). \quad (32.2.4)$$

If in addition the observations fall into g clusters with $\mathbf{R} = \bigoplus_k \mathbf{R}_k$, $\mathbf{Z} = \bigoplus_k \mathbf{Z}_k$ and $\mathbf{G} = \bigoplus_k \mathbf{G}_k$ block diagonal, the last saying that different clusters' random effects are independent, then $\mathbf{V} = \bigoplus_k (\mathbf{Z}_k \mathbf{G}_k \mathbf{Z}_k^\top + \mathbf{R}_k)$ is block diagonal too, and everything decomposes cluster by cluster.

Proof. $\mathbf{V} = \mathbf{R} + \mathbf{Z}\mathbf{G}\mathbf{Z}^\top$ is a sum of a positive definite and a nonnegative definite matrix, hence positive definite. Now use Theorem 1.6.5 with the roles $\mathbf{B} \mapsto \mathbf{R}$, $\mathbf{U} \mapsto \mathbf{Z}$, $\mathbf{C} \mapsto \mathbf{G}$ and (in the notation of that theorem) the second outer factor also $\mathbf{Z}$. The matrix $\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z}$ is positive definite, hence nonsingular, so the hypothesis holds and Eq. 1.6.6 is [Eq. 32.2.3](#). For the determinant, Theorem 1.6.3(c) with $\mathbf{B} = \mathbf{R}$, $\mathbf{U} = \mathbf{Z}\mathbf{G}$ and the second factor $\mathbf{Z}^\top$ gives

$$\begin{aligned} \det \mathbf{V} &= \det \mathbf{R} \det(\mathbf{I}_q + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z}\mathbf{G}) \\ &= \det \mathbf{R} \det \mathbf{G} \det(\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z}), \end{aligned}$$

the last step by $\det(\mathbf{I}_q + \mathbf{A}\mathbf{G}) = \det(\mathbf{G}^{-1} + \mathbf{A}) \det \mathbf{G}$. The block diagonal statement is immediate from the definitions. ■

A study of $n = 10^5$ pupils in $q = 500$ schools therefore needs a 500×500 inverse, not a $10^5 \times 10^5$ one, and if $\mathbf{R}$ is diagonal the rest of [Eq. 32.2.3](#) costs nothing. [Section 32.3](#) improves on this again.

32.2.3 Four examples

Example. Random intercepts

Observations are grouped, group k contributing n_k of them. Take $\mathbf{Z}$ to be the $n \times g$ matrix of group indicators, $\mathbf{G} = \sigma_a^2 \mathbf{I}_g$ and $\mathbf{R} = \sigma^2 \mathbf{I}_n$, so that $\mathbf{V} = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$ as in Propo-

sition 32.1.1(a). The fixed part $\mathbf{X}\boldsymbol{\beta}$ may hold anything: an intercept alone, recovering Definition 32.1.1; a treatment indicator constant within group, as in a cluster-randomized trial; or regressors that vary within a group, as in a longitudinal study. Each group's block of $\mathbf{V}$ is $\sigma^2\mathbf{I}_{n_k} + \sigma_a^2\mathbf{1}\mathbf{1}^\top$, the **compound symmetric** or **equicorrelated** form.

Example. Random intercepts and slopes

Subject i is measured at times $t_{i1}, \dots, t_{in_i}$, and each subject has its own intercept and its own slope in time:

$$Y_{ij} = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + u_{0i} + u_{1i}t_{ij} + \varepsilon_{ij}, \quad \begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim N_2(\mathbf{0}, \mathbf{D}),$$

independently across subjects, with $\varepsilon_i \sim N_{n_i}(\mathbf{0}, \sigma^2\mathbf{I})$. Here $\mathbf{Z}_i = [\mathbf{1}, \mathbf{t}_i]$, $\mathbf{G} = \mathbf{I}_N \otimes \mathbf{D}$ for N subjects, and subject i 's covariance block is $\mathbf{V}_i = \mathbf{Z}_i\mathbf{D}\mathbf{Z}_i^\top + \sigma^2\mathbf{I}$, with entries

$$\text{Cov}(Y_{ij}, Y_{ij'}) = d_{11} + d_{12}(t_{ij} + t_{ij'}) + d_{22}t_{ij}t_{ij'} + \sigma^2\mathbf{1}\{j = j'\}.$$

Two features deserve notice. The variance of a response is quadratic in time, so a random slope model necessarily says that subjects spread apart, or, if $d_{12} < 0$, first converge and then spread. And the correlation between two occasions depends on both times, not only on their separation: it is not stationary. Whether this is the right shape is a modelling question that [Chapter 33](#) takes up.

Example. Nested and crossed random factors

With pupils inside classrooms inside schools, write $\mathbf{Z} = [\mathbf{Z}_{\text{sch}}, \mathbf{Z}_{\text{cls}}]$ and $\mathbf{G} = \begin{pmatrix} \sigma_S^2\mathbf{I} & \mathbf{0} \\ \mathbf{0} & \sigma_C^2\mathbf{I} \end{pmatrix}$. Because every classroom belongs to exactly one school, $\mathbf{V}$ is block diagonal by school, and the analysis splits school by school; this is the covariance behind Exercise (C1). With subjects responding to items, each subject seeing every item, the two factors are **crossed**: $\mathbf{Z}_{\text{sub}}$ and $\mathbf{Z}_{\text{item}}$ generate overlapping patterns and $\mathbf{V}$ has no block diagonal form at all. Nothing in the theory changes, but the computation does, and crossed random effects are the reason modern software stores $\mathbf{Z}$ sparsely.

Example. Grunfeld's panel with a random firm effect

[Section 6.6](#) fitted gross investment on market value and capital stock for 11 US firms over 20 years, once with a separate intercept per firm (the within estimator) and once with a single intercept. The mixed model puts the firm effect in between: $\mathbf{X}$ holds the intercept, value and capital, $\mathbf{Z}$ holds the firm indicators, $\mathbf{G} = \sigma_f^2\mathbf{I}_{11}$ and $\mathbf{R} = \sigma^2\mathbf{I}$. Fitted by restricted maximum likelihood ([Section 32.4](#)), it gives $\hat{\sigma}^2 = 2527.6$ and $\hat{\sigma}_f^2 = 6740.9$, a firm standard deviation of 82.1 and an intraclass correlation of 0.727: most of the variation in investment is between firms. The coefficients are 0.1094 for value and 0.3082 for capital, against 0.1101 and 0.3100 for the within estimator and 0.1145 and 0.2275 with a single intercept. The mixed estimate has landed essentially on the within estimate, and [Section 32.3](#) explains why: with 20 years per firm, the weight the model gives to between-firm comparisons is $1 - 0.9816$, less than two per cent.

```
import numpy as np
```

```

import pandas as pd
import statsmodels.api as sm

df = sm.datasets.grunfeld.load_pandas().data.sort_values(["firm", "year"])
firms = df["firm"].unique()
y = df["invest"].to_numpy()
X = np.column_stack([np.ones(len(y)), df["value"], df["capital"]])
blocks = [np.where(df["firm"].to_numpy() == f)[0] for f in firms]
n, p, g = len(y), X.shape[1], len(firms)
print(f"{n} observations, {g} firms, {n // g} years each")

```

[Run this code online](#)

32.2.4 When ordinary least squares is already best

The marginal model [Eq. 32.2.2](#) is a general Gauss–Markov model, so [Corollary 7.2.3](#) says the best linear unbiased estimator of an estimable $\lambda^\top \beta$ is the generalized least squares estimate built from $\mathbf{V}^{-1}$, and [Theorem 31.2.2](#) says that ordinary least squares achieves this for every $\mathbf{y}$ exactly when $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$. Mixed models supply the standard examples on both sides of that line.

Example. Cluster-constant regressors

Take [Example \(Random intercepts\)](#) with a balanced layout, $n_k = m$, and suppose every column of $\mathbf{X}$ is constant within groups, so that $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$. By [Eq. 32.1.2](#), $\mathbf{V} = \lambda_1 \mathbf{M} + \sigma^2(\mathbf{I} - \mathbf{M})$ with $\mathbf{M}$ the projection onto $\mathcal{C}(\mathbf{Z})$, so $\mathbf{V}\mathbf{X} = \lambda_1 \mathbf{X}$ and $\mathcal{C}(\mathbf{V}\mathbf{X}) = \mathcal{C}(\mathbf{X})$. Ordinary least squares is therefore the best linear unbiased estimator, whatever the variance components are, and the analysis of a balanced cluster-randomized trial can be carried out on the cluster means without loss. The covariance of the estimate is *not* $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$, however: it is $\lambda_1(\mathbf{X}^\top \mathbf{X})^{-1}$, larger by the factor $1 + m\gamma$. Getting the point estimate right and the standard error wrong is the classic error in clustered data.

Example. Regressors that vary within a cluster

Keep the same $\mathbf{V}$, but let $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\mathbf{x}$ varying within groups. Decomposing $\mathbf{x}$ into its between and within parts,

$$\mathbf{V}\mathbf{x} = \lambda_1 \mathbf{M}\mathbf{x} + \sigma^2(\mathbf{I} - \mathbf{M})\mathbf{x},$$

which lies in $\mathcal{C}(\mathbf{X})$ only if one of the two parts vanishes or $\lambda_1 = \sigma^2$. Ordinary least squares is then not efficient, and the reason is visible: generalized least squares downweights the between-group part of $\mathbf{x}$ by $\sigma^2/\lambda_1 = 1/(1 + m\gamma)$ relative to the within-group part, because between-group comparisons carry the extra variance σ_a^2 . The mixed estimate is a compromise between the within estimator of [Section 6.6](#), which uses only $(\mathbf{I} - \mathbf{M})\mathbf{x}$, and the between estimator, which uses only $\mathbf{M}\mathbf{x}$; as $m\gamma \rightarrow \infty$ the weight on the between part goes to zero. [Example \(Grunfeld’s panel with a random firm effect\)](#) is that limit in practice.

Idea. A random effect is a statement about *where the information is*. Declaring the grouping factor random says that comparisons between groups carry an extra variance σ_a^2 and should be downweighted accordingly, but not discarded. Fixed effects for the groups discard them

entirely; ignoring the grouping keeps them at full weight. The mixed model interpolates, and the interpolation weight is estimated from the data.

The remaining difficulty is that $\mathbf{V}$ depends on the unknown $\boldsymbol{\theta}$. The estimator actually used is the feasible generalized least squares estimator of Definition 31.4.1,

$$\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}) = (\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{Y}, \quad (32.2.5)$$

with $\hat{\boldsymbol{\theta}}$ from Section 32.4. Theorem 31.4.1 gives the large-sample justification; Section 32.5 asks what it costs in small samples, and the answer is: more than people usually allow for.

32.2.5 Exercises

32.2.5.1 A. Check your understanding

Exercise. A1

A study measures 3 subjects on 2 occasions each, with a random subject effect of variance σ_s^2 and independent errors of variance σ^2 . Write $\mathbf{Z}$, $\mathbf{G}$, $\mathbf{R}$ and $\mathbf{V}$ explicitly as 6×6 or 6×3 matrices, and check Eq. 32.2.3 numerically for $\sigma_s^2 = 2$, $\sigma^2 = 1$.

Exercise. A2

A colleague argues that in Example (Random intercepts) the two variance components cannot be separated, since raising σ_a^2 and lowering σ^2 both change the diagonal of $\mathbf{V}$ in the same direction. Is the argument right? What happens if every group has $n_k = 1$?

32.2.5.2 B. Practice

Exercise. B1

Specialize Eq. 32.2.3 to Example (Random intercepts) with group sizes n_k , and show that the k th block of $\mathbf{V}^{-1}$ is

$$\frac{1}{\sigma^2} \left(\mathbf{I}_{n_k} - \frac{\sigma_a^2}{\sigma^2 + n_k \sigma_a^2} \mathbf{1} \mathbf{1}^\top \right).$$

Deduce that the generalized least squares criterion is $\sigma^{-2} \sum_k \{ \sum_j r_{kj}^2 - w_k n_k \bar{r}_k^2 \}$ with $w_k = n_k \sigma_a^2 / (\sigma^2 + n_k \sigma_a^2)$, and interpret w_k .

Solution. Each block of $\mathbf{V}$ is $\sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{1} \mathbf{1}^\top$; apply Eq. 1.6.7 with $\mathbf{B} = \sigma^2 \mathbf{I}$, $\mathbf{u} = \mathbf{v} = \sigma_a \mathbf{1}$: the correction is $\sigma_a^2 \sigma^{-4} \mathbf{1} \mathbf{1}^\top / (1 + n_k \sigma_a^2 / \sigma^2)$, which rearranges to the stated form. The criterion $\mathbf{r}^\top \mathbf{V}^{-1} \mathbf{r}$ then equals $\sigma^{-2} \sum_k \{ \sum_j r_{kj}^2 - n_k \bar{r}_k^2 w_k \}$ after writing $\mathbf{1}^\top \mathbf{r}_k = n_k \bar{r}_k$. The weight $w_k \in (0, 1)$ increases with n_k : the model removes a fraction w_k of the group's mean residual from the criterion, because that part of the residual is explained by the group's random effect rather than counted against the fit.

Exercise. B2

Let $\mathbf{V} = \lambda_1 \mathbf{M} + \sigma^2(\mathbf{I} - \mathbf{M})$ as in Eq. 32.1.2. Show that $\mathcal{C}(\mathbf{VX}) \subseteq \mathcal{C}(\mathbf{X})$ holds if and only if $\mathcal{C}(\mathbf{X})$ decomposes as $(\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{Z})) \oplus (\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{Z})^\perp)$, that is, iff $\mathbf{M}$ maps $\mathcal{C}(\mathbf{X})$ into itself. Give an $\mathbf{X}$ with both between and within columns for which ordinary least squares *is* best.

Solution. $\mathbf{VX} = \sigma^2 \mathbf{X} + (\lambda_1 - \sigma^2) \mathbf{MX}$, so (assuming $\lambda_1 \neq \sigma^2$) $\mathcal{C}(\mathbf{VX}) \subseteq \mathcal{C}(\mathbf{X})$ iff $\mathcal{C}(\mathbf{MX}) \subseteq \mathcal{C}(\mathbf{X})$, that is, $\mathbf{M}$ maps $\mathcal{C}(\mathbf{X})$ into itself. A subspace invariant under an orthogonal projection splits into its parts inside $\mathcal{C}(\mathbf{Z})$ and inside $\mathcal{C}(\mathbf{Z})^\perp$. Example: $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ with $\mathbf{1}$ constant within groups and $\mathbf{x}$ having zero mean *within every group*, so $\mathbf{Mx} = \mathbf{0}$; then $\mathbf{MX} = [\mathbf{1}, \mathbf{0}]$ lies in $\mathcal{C}(\mathbf{X})$. A balanced split-plot design has exactly this structure, which is why [Chapter 33](#) can analyse it stratum by stratum.

32.2.5.3 C. Going deeper**Exercise. C1**

Two crossed random factors with a and b levels and one observation per cell give $\mathbf{Z} = [\mathbf{Z}_A, \mathbf{Z}_B]$ with $n = ab$, $q = a + b$. Show that $\text{rank}(\mathbf{Z}) = a + b - 1$, so $\mathbf{Z}^\top \mathbf{Z}$ is singular, and explain why this causes no difficulty for [Eq. 32.2.2](#) but would if one tried to treat $\mathbf{u}$ as a parameter vector to be estimated. Compute $\mathbf{V}$ and show it is *not* block diagonal under any ordering of the observations when $a, b \geq 2$.